AIORI 3 HACKATHON PROJECT · A3-PS026-TC249

AT URI Syntax Compatibility and Round-Trip Validator

PROPOSED SOLUTION AND USER WORKFLOW

1. ENTER AN AT URI THROUGH THE WEBSITE

2. PROCESS THE INPUT THROUGH THE VALIDATION PIPELINE

3. RUN AT URI SYNTAX VALIDATION

4. EVALUATE COMPATIBILITY WITH RFC 3986 SYNTAX

5. IDENTIFY INVALID SYNTAX AND EDGE CASES

6. DISPLAY A CLEAR VALIDATION RESULT

INNOVATION

UNIFIED INTERFACE

Web-based AT URI validation in one place.

DIRECT COMPARISON

AT URI requirements are compared with RFC 3986 generic syntax.

EXPLICIT CLASSIFICATION

Inputs are classified as valid, invalid, or edge-case.

105 TEST CASES

Predefined cases support systematic validation.

AUTOMATED EVALUATION

Less reliance on manual inspection.

CONSISTENT PARSING

Focus on compatibility and repeatable behaviour.

SYSTEM ARCHITECTURE

1. USER → WEB INTERFACE
2. URI INPUT PROCESSING
3. AT URI SYNTAX
4. RFC 3986 COMPATIBILITY PARSER
5. COMPARISON AND VALIDATION ENGINE
6. TEST CASE EVALUATION · 105 TEST CASES
7. RESULT GENERATION
8. VALID · INVALID · EDGE CASE

COMPONENT RESPONSIBILITIES

Component	Responsibility
Web interface	Accept manual URI input and present results.
Validators and parser	Apply AT URI rules and RFC 3986 compatibility checks.
Comparison engine	Compare outcomes and identify discrepancies.
Test evaluator	Run the predefined test set consistently.
Result generation	Translate findings into clear classifications.

OUTCOME

Demonstrated or intended capability	Status framing
Manual URI validation through a website	**Core demonstrated functionality**
Valid, invalid, and edge-case identification	Core validation outcome
AT URI versus RFC 3986 comparison	Core compatibility analysis
Evaluation with 105 predefined test cases	Systematic test scope
Clear presentation of results	User-facing output goal